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AI/ML Team – Member 2 Report

Body Segmentation and Human Parsing for Virtual Try-On Systems

Introduction

Virtual Try-On systems allow users to digitally visualize clothing on their bodies before purchasing. One of the most important preprocessing steps in these systems is Body Segmentation and Human Parsing. These techniques help identify the person in an image, separate them from the background, and detect various body regions for accurate clothing placement.

Body segmentation focuses on isolating the human body from the surrounding environment, while human parsing provides a more detailed understanding of different body parts and clothing regions. These technologies play a significant role in improving the accuracy and realism of virtual try-on applications.

This report explores body segmentation concepts, human parsing techniques, and the MediaPipe Selfie Segmentation model used for experimentation.

Body Segmentation

Body segmentation is a computer vision technique that separates a person from the background in an image or video. The output is generally a segmentation mask that identifies which pixels belong to the human body.

In virtual try-on applications, body segmentation helps:

- Remove unwanted background information.
- Identify the user's body shape.
- Improve garment alignment.
- Enhance virtual clothing fitting accuracy.

Body segmentation serves as the foundation for many advanced fashion AI applications.

Human Parsing

Human parsing is a more detailed version of segmentation. Instead of identifying only the person, it classifies different body parts and clothing regions separately.

Examples of body regions detected during human parsing include:

- Hair
- Face

- Arms
- Legs
- Torso
- Upper Clothing
- Lower Clothing
- Shoes

Human parsing enables virtual try-on systems to place garments more accurately by understanding the structure of the human body.

MediaPipe Selfie Segmentation

MediaPipe Selfie Segmentation is a lightweight segmentation framework developed for real-time human segmentation. It can efficiently separate a person from the background using a webcam or image input.

The model is optimized for speed and can run on desktop systems, mobile devices, and web applications.

Features

- Real-time processing.
- Fast segmentation performance.
- Lightweight architecture.
- Easy implementation.

Advantages

- Low computational requirements.
- Quick response time.
- Suitable for live camera applications.
- Simple integration into AI projects.

Limitations

- Minor inaccuracies around hair boundaries.
- Reduced precision in complex backgrounds.
- Less detailed compared to advanced human parsing models.

Experiment Setup

The MediaPipe Selfie Segmentation model was tested using a webcam feed. The system captured live video frames and generated segmentation results in real time.

The experiment focused on observing:

- Segmentation quality.
- Background removal performance.

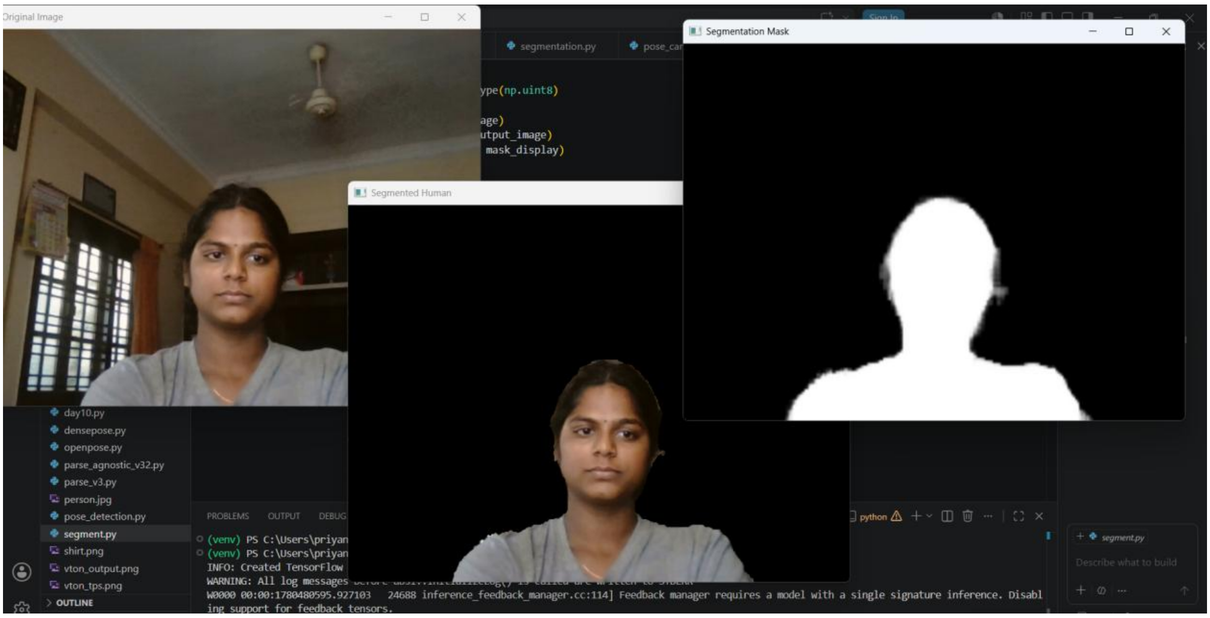
- Processing speed.
- Edge accuracy around the body.

Experimental Results

The original webcam image was captured before applying segmentation. This image contained both the person and the surrounding background.

The segmentation model successfully separated the person from the background. Most body regions were preserved while background pixels were removed.

A binary segmentation mask was generated where the human body was highlighted and the background was suppressed. This mask can be used as input for virtual try-on systems.



Observations

The MediaPipe Selfie Segmentation model demonstrated fast and efficient performance during testing. The model was able to process frames in real time while maintaining good segmentation quality.

Observations obtained during the experiment include:

Parameter	**Observation**
Processing Speed	Very Fast
Background Removal	Good
Segmentation Accuracy	Good
Edge Detection	Moderate

Parameter	Observation
Hair Segmentation	Minor inaccuracies
Real-Time Performance	Excellent
Virtual Try-On Suitability	Suitable for preprocessing

Applications in Virtual Try-On Systems

Body segmentation and human parsing are widely used in modern fashion AI systems. Their applications include:

- Virtual fitting rooms.
- Online fashion shopping platforms.
- Garment visualization systems.
- Personalized clothing recommendations.
- Augmented reality fashion applications.

By accurately identifying body regions, these techniques help improve clothing placement and overall user experience.

Findings

The experiment demonstrated that MediaPipe Selfie Segmentation provides an effective solution for real-time body segmentation. The model successfully removed backgrounds and generated usable segmentation masks suitable for virtual try-on preprocessing tasks.

Although more advanced models such as DeepLabV3, SCHP, and Detectron2 provide greater accuracy, MediaPipe offers an excellent balance between speed and performance, making it suitable for real-time applications.

Conclusion

Body Segmentation and Human Parsing are essential components of virtual try-on systems. They enable accurate identification of human body regions and improve clothing alignment during digital garment fitting.

The MediaPipe Selfie Segmentation model successfully performed real-time segmentation and produced reliable results during experimentation. The generated segmentation masks can be effectively used in virtual try-on pipelines, making MediaPipe a practical solution for fashion AI applications.

Future work may involve evaluating advanced human parsing models such as DeepLabV3, SCHP, and Detectron2 to achieve even greater segmentation accuracy and detailed body-part recognition.